

Data Structures

Laboratory Sheet No. 4

Engineering - Instituto Superior de Engenharia de Coimbra

- 1 - Build a `TenDoubles` class that allows the user to store 10 numbers (`Double` objects). Numbers can be added using the `add` method (`double`). You can not remove numbers. If the user attempts to insert numbers beyond the limit, an exception appropriate `RuntimeException` should be generated.
- 2 - Build a suitable iterator class for `TenDoubles`, modifying it so that it now implements the interface `Iterable <Double>`. Ignore for now the possibility of exceptions occur beyond `UnsupportedOperationException`.
- 3 - Add the iterator built in the previous question support for generating appropriate `NoSuchElementException` exceptions.
- 4 - Create a `TenDoublesMutable` class similar to `TenDoubles` that allows removal elements. Create an appropriate iterator, ensuring the correct generation of `IllegalStateException` and `NoSuchElementException` exceptions.
- 5 - Add the necessary support for the iterator throwing the `ConcurrentModificationException` if the data container is changed externally.
- 6 - Create an alternative iterator that traverses only the positive elements (this should become the iterator by default).
- 8 - Build an algorithm that searches for the largest number in a set of iterable `Doubles`. Make sure your algorithm works properly with `TenDoubles` class `TenDoublesMutable` and the class `ArrayList<Double>`.
- 9 - Generalize the previous algorithm in a way that works for any collection with comparable features